



Commonwealth of Virginia
VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

PIEDMONT REGIONAL OFFICE
4949-A Cox Road, Glen Allen, Virginia 23060
Phone: (804) 527-5020 Fax: (804) 698-4178
www.deq.virginia.gov

Matthew J. Strickler
Secretary of Natural Resources

David K. Paylor
Director
(804) 698-4000

James J. Golden
Regional Director

February 9, 2021

Ms. Margaret Sexton
Sr. HSE Director
CBRE / Bank of America Account
525 N. Tyron St.
NC1-023-05-55
Charlotte, NC 28255

Location: Henrico County
Registration No.: 52403

Dear Ms. Sexton,

Attached is a permit to construct and operate a project at a banking and data center in accordance with the provisions of the Virginia State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. This permit supersedes your permit dated January 13, 2020.

In the course of evaluating the application and arriving at a final decision to approve the project, the Department of Environmental Quality (DEQ) deemed the application complete on January 29, 2021.

This permit contains legally enforceable conditions. Failure to comply may result in a Notice of Violation and/or civil charges. Please read all permit conditions carefully.

This permit approval to construct and operate shall not relieve Bank of America National Association of the responsibility to comply with all other local, state, and federal permit regulations.

The generators (Ref. Nos. 1-5, 12, 14, 15, 18-21, 26-31, 34-39, and 42-47) are subject to 40 CFR 63, Maximum Achievable Control Technology, (MACT) Subpart ZZZZ and 40 CFR 60, New Source Performance Standard (NSPS), Subpart IIII. Virginia has not accepted delegation of these rules for minor sources. In summary, the units are required to comply with certain federal emission standards and operating limitations. DEQ advises you to review the referenced MACT and NSPS to ensure compliance with applicable emission and operational limitations. As the owner/operator you are also responsible for any monitoring, notification, reporting and recordkeeping requirements of the MACT and NSPS. Notifications shall only be sent to EPA, Region III.

To review any federal rules referenced in the above paragraph or in the attached permit, the US Government Publishing Office maintains the text of these rules at www.ecfr.gov, Title 40, Parts 60 and 63.

The Board's Regulations as contained in Title 9 of the Virginia Administrative Code 5-170-200 provide that you may request a formal hearing from this case decision by filing a petition with the Board within 30 days after this case decision notice was mailed or delivered to you. Please consult the relevant regulations for additional requirements for such requests.

As provided by Rule 2A:2 of the Supreme Court of Virginia, you have 30 days from the date you actually received this permit or the date on which it was mailed to you, whichever occurred first, within which to initiate an appeal of this decision by filing a Notice of Appeal with:

David K. Paylor, Director
Department of Environmental Quality
P. O. Box 1105
Richmond, VA 23218

If this permit was delivered to you by mail, three days are added to the thirty-day period in which to file an appeal. Please refer to Part Two A of the Rules of the Supreme Court of Virginia for information on the required content of the Notice of Appeal and for additional requirements governing appeals from decisions of administrative agencies.

If you have any questions concerning this permit, please contact the regional office at 804-527-5020.

Sincerely,

A handwritten signature in blue ink, appearing to read 'J. Kyle', with a horizontal line extending to the right.

James E. Kyle, P.E.
Air Permit Manager

JEK/AMS/52403_09_2021 NSR permit .docx

Attachments: Permit

cc: PRO Air Compliance



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Regional Director

STATIONARY SOURCE PERMIT TO CONSTRUCT AND OPERATE

This permit supersedes your permit dated January 13, 2020.

In compliance with the Federal Clean Air Act and the Commonwealth of Virginia Regulations for the Control and Abatement of Air Pollution,

Bank of America National Association
6050 Technology Creek Drive
Sandston, Virginia 23150
Registration No.: 52403

is authorized to construct and operate

a data center

located at

6050 Technology Creek Drive
Sandston, Virginia 23150

in accordance with the Conditions of this permit.

Approved on February 9, 2021.

A handwritten signature in blue ink, appearing to read "J. Kyle", written over a horizontal line.

James E. Kyle, P.E.
Air Permit Manager

Permit consists of 10 pages.
Permit Conditions 1 to 25.

INTRODUCTION

This permit approval is based on the permit applications dated December 2, 2011; July 20, 2012; August 30, 2016; June 6, 2017; September 1, 2017; January 15, 2019; November 8, 2019; and January 25, 2021, including amendment information dated January 9, 2012, April 2, 2012, April 13, 2012, August 13, 2012, September 11, 2012, and October 10, 2016 and supplemental information dated July 6 and 27, 2017, March 11, 2019 and December 17, 2019. Any changes in the permit application specifications or any existing facilities which alter the impact of the facility on air quality may require a permit. Failure to obtain such a permit prior to construction may result in enforcement action. In addition, this facility may be subject to additional applicable requirements not listed in this permit.

Words or terms used in this permit shall have meanings as provided in 9 VAC 5-10-10 of the State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. The regulatory reference or authority for each condition is listed in parentheses () after each condition.

Annual requirements to fulfill legal obligations to maintain current stationary source emissions data will necessitate a prompt response by the permittee to requests by the DEQ or the Board for information to include, as appropriate: process and production data; changes in control equipment; and operating schedules. Such requests for information from the DEQ will either be in writing or by personal contact.

The availability of information submitted to the DEQ or the Board will be governed by applicable provisions of the Freedom of Information Act, §§ 2.2-3700 through 2.2-3714 of the Code of Virginia, § 10.1-1314 (addressing information provided to the Board) of the Code of Virginia, and 9 VAC 5-170-60 of the State Air Pollution Control Board Regulations. Information provided to federal officials is subject to appropriate federal law and regulations governing confidentiality of such information.

Equipment List - Equipment at this facility covered by this permit consists of:
Equipment included in the project:

Reference No.	Equipment Description (model year)	Rated Capacity	Delegated Federal Requirements
42-47	Six Evolve diesel emergency engine-generators, engine Model 6135HFG75 (2020)	563 kW/755 hp (engine rating), each	None

Other permitted equipment:

Reference No.	Equipment Description (model year)	Rated Capacity	Delegated Federal Requirements
1-5	Five Cummins Model DQKAC diesel engine-generator, engine model QSK60-G14 NR2 (2012)	2,446 kW/3280 hp (engine rating), each	None
14, 15 and 18-21	Six Caterpillar diesel emergency engine-generators, Model 3512C, 14-15 (2016), 18-21 (2017)	1,645 kW/2206 hp (engine rating), each	None
26-31 and 34-39	Twelve Clarke diesel emergency engine-generators, engine Model 6135HFG75, 26-29 (2018), 30-31 (2019) and 34-39 (2020)	563 kW/755 hp (engine rating), each	None

Equipment Exempt from Permitting:

Reference No.	Equipment Description	Rated Capacity	Exemption Citation 9 VAC 5-80-
6 and 7	Two Lochinvar Crest FBN3000 natural gas boilers	3.0 MMBtu/hr, each	1105 B.1
8 and 9	Two Pressure storage tanks (diesel)	33,000 gal, each	1105 B.8.e
10	Pressure storage tank (diesel)	15,000 gal	1105 B.8.e
11	Pressure storage tank (urea)	15,000 gal	1105
12	Cummins diesel emergency generator	250 kW	1105 B.2.b
13	Pressure storage tank (diesel)	300 gal	1105 B.8.e
16 and 17	Two Storage tanks (diesel)	5,900 gal, each	1105 B.8.e
22-25	Four Storage tanks (diesel)	5,800 gal, each	1105 B.8.e
32, 33, 40, 41, 48, and 49	Six Storage tanks (diesel)	5,000 gal, each	1105 B.8.e

Specifications included in the above tables are for informational purposes only and do not form enforceable terms or conditions of the permit.

PROCESS REQUIREMENTS

- Emissions Controls** – Visible emissions, particulate emissions, carbon monoxide emissions, volatile organic compound emissions and nitrogen oxide emissions shall be controlled by the use of good operating practices and performing appropriate maintenance in accordance with the manufacturer recommendations. In addition, the permittee may only change those settings that are permitted by the manufacturer which do not increase air emissions.
(9 VAC 5-80-1180)
- Emission Controls** – Nitrogen oxides (NO_x) emissions from the engine-generator sets (Ref. Nos. 14, 15, 18 – 21, 26-31, 34-39, and 42-47) shall be controlled by the following: electronic fuel injection and/or turbocharged engine and/or aftercooler and/or charge air cooler and/or low NO_x emission package. The permittee shall maintain documentation that demonstrates the control devices have been installed on the engine-generator sets.
(9 VAC 5-80-1180 and 9 VAC 5-50-260)
- Monitoring Devices** – Each engine-generator set (Ref. Nos. 1-5, 14, 15, 18 -21, 26-31, 34-39, and 42-47) shall be equipped with non-resettable hour metering device to monitor the operating hours. The non-resettable hour meters used to continuously measure the hours of operation for each engine-generator set shall be observed by the owner with a frequency of not less than once each day the engine-generator set is operated. The owner shall keep a log of these observations. Each monitoring device shall be installed, maintained, calibrated (as appropriate) and operated in accordance with approved procedures that shall include, as a minimum, the manufacturer's written requirements or recommendations. Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the generators are operating.
(9 VAC 5-80-1180 D)

OPERATING LIMITATIONS

4. **Operation of the Engine-Generator Set** - The permittee shall operate and maintain each engine-generator set and control device according to the manufacturer's written instructions or procedures developed by the permittee that are approved by the engine manufacturer. In addition, the permittee may only change those settings that are permitted by the manufacturer which do not increase air emissions.
(9 VAC 5-80-1180)
5. **Operating Hours** – Each engine-generator set (Ref. Nos. 1-5), shall not operate more than 1000 hours per year, calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
(9 VAC 5-80-1180 and 9 VAC 5-50-260)
6. **Operating Hours** – Each engine-generator set (Ref. Nos. 14, 15 and 18-21), shall not operate more than 500 hours per year and all engine-generator sets referenced in this condition shall not operate more than 2,130 hours per year combined, calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
(9 VAC 5-80-1180)
7. **Operating Hours** – Each engine-generator set (Ref. Nos. 26-31, 34-39, 42-47), shall not operate more than 500 hours per year and all engine-generator sets referenced in this condition shall not operate more than 6,390 hours per year combined, calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
(9 VAC 5-80-1180)
8. **Fuel** - The approved fuel for the engine-generator sets (Ref. Nos. 1-5, 14, 15, 18-21, 26-31 34-39, and 42-47) is diesel fuel. The diesel fuel shall meet the ASTM D975 specification for S15 diesel fuel oil with a maximum sulfur content per shipment of 0.0015%. A change in the fuel shall be considered a change in the method of operation of the engine-generator sets and may require a new or amended permit. However, if a change in the fuel is not subject to new source review permitting requirements, this condition should not be construed to prohibit such a change.
(9 VAC 5-80-1180)
9. **Fuel Certification** - The permittee shall obtain a certification from the fuel supplier with each shipment of diesel fuel. Each fuel supplier certification shall include the following:
 - a. The name of the fuel supplier;

- b. The date on which the diesel fuel was received;
- c. The quantity of diesel fuel delivered in the shipment;
- d. A statement that the diesel fuel complies with the American Society for Testing and Materials specifications (ASTM D975) for S15 diesel fuel oil; and
- e. The sulfur content of the diesel fuel.

Fuel sampling and analysis, independent of that used for certification, as may be periodically required or conducted by DEQ may be used to determine compliance with the fuel specifications stipulated in Condition 8. Exceedance of these specifications may be considered credible evidence of the exceedance of emission limits.
(9 VAC 5-80-1180)

EMISSION LIMITS

10. **Process Emission Limits** - Emissions from the operation of the engine-generator sets (Ref. Nos. 1-5) shall not exceed the limits specified below:

	Each Unit	Combined Units
PM	0.7 lb/hr	1.7 tons/yr
PM10	0.7 lb/hr	1.7 tons/yr
Nitrogen Oxides (as NO ₂)	5.4 lb/hr	13.5 tons/yr
Carbon Monoxide	18.9 lb/hr	47.2 tons/yr
Volatile Organic Compounds	2.2 lb/hr	5.4 tons/yr

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with these emission limits may be determined as stated in Conditions 1, 4, 5, and 8.
(9 VAC 5-80-1180 and 9 VAC 5-50-260)

11. **Process Emission Limits** - Emissions from the operation of the engine-generator sets (Ref. Nos. 14, 15, 18- 21) shall not exceed the limits specified below:

	Each Unit	Combined Units
Nitrogen Oxides (as NO ₂)	6.0 g/hp-hr	33.7 tons/yr
Carbon Monoxide	4.3 lb/hr	4.6 tons/yr
Volatile Organic Compounds	1.4 lb/hr	1.5 tons/yr

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the

exceedance of emission limits. Compliance with these emission limits may be determined as stated in Conditions 1, 2, 4, 6, and 8.

(9 VAC 5-80-1180 and 9 VAC 5-50-260)

12. **Process Emission Limits** - Emissions from the operation of the engine-generator sets (Ref. Nos. 26-31, 34-39, and 42-47) shall not exceed the limits specified below:

	Each Unit	Combined Units
Nitrogen Oxides (as NO ₂)	4.9 g/hp-hr	26.3 tons/yr
Carbon Monoxide	0.8 lb/hr	2.6 tons/yr
Volatile Organic Compounds	0.5 lb/hr	1.6 tons/yr

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with these emission limits may be determined as stated in Conditions 1, 2, 4, 7, and 8.

(9 VAC 5-80-1180 and 9 VAC 5-50-260)

13. **Visible Emission Limit** - Visible emissions from each engine-generator set exhaust (Ref. Nos. 1-5, 14, 15, 18-21, 26-31, 34-39, and 42-47) each shall not exceed 10% opacity except during one six-minute period in any one hour in which visible emissions shall not exceed 20% opacity as determined by the EPA Method 9 (reference 40 CFR 60, Appendix A). This condition applies at all times except during startup, shutdown, and malfunction.

(9 VAC 5-80-1180)

RECORDS

14. **On Site Records** - The permittee shall maintain records of emission data and operating parameters as necessary to demonstrate compliance with this permit. The content and format of such records shall be arranged with the Piedmont Regional Office. These records shall include, but are not limited to:

- Annual hours of operation for each engine-generator set, calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
- All fuel supplier certifications.
- Engine information including make, model, serial number, model year, maximum engine power (bhp), and engine displacement for each engine-generator set.
- The manufacturer's written operating instructions or procedures developed by the owner/operator that are approved by the engine manufacturer for each engine-generator set.

- e. Records of the reasons for operation for each engine-generator set (Ref. Nos. 14, 15, 18-21, 26-31, 34-39, and 42-47), including, but not limited to, the date, and cause of operation.
- f. Scheduled and unscheduled maintenance and operator training.
- g. Results of all stack tests, visible emission evaluations and performance evaluations.

These records shall be available for inspection by the DEQ and shall be current for the most recent five years.

(9 VAC 5-80-1180 and 9 VAC 5-50-50)

PROCESS REQUIREMENTS

15. **Emissions Testing** - The facility shall be constructed so as to allow for emissions testing upon reasonable notice at any time, using appropriate methods. Sampling ports shall be provided when requested at the appropriate locations and safe sampling platforms and access shall be provided.

(9 VAC 5-50-30 F and 9 VAC 5-80-1180)

NOTIFICATIONS

16. **Initial Notifications** - The permittee shall furnish written notification to the Piedmont Regional Office of:

- a. The actual date on which construction of the engine-generator sets (Ref. Nos. 42-47) commenced within 30 days after such date.
- b. The anticipated start-up date of the engine-generator sets (Ref. Nos. 42-47) postmarked not more than 60 days nor less than 30 days prior to such date.
- c. The actual start-up date of the engine-generator sets (Ref. Nos. 42-47) within 15 days after such date.

(9 VAC 5-50-50 and 9 VAC 5-80-1180)

GENERAL CONDITIONS

17. **Permit Invalidation** – This permit to construct the engine-generator sets (Ref. Nos. 42-47) shall become invalid, unless an extension is granted by the DEQ, if:

- a. A program of continuous construction is not commenced within 18 months from the date of this permit.

- b. A program of construction or modification is discontinued for a period of 18 months or more, or is not completed within a reasonable time, except for a DEQ approved period between phases of the phased construction of a new stationary source or project.

(9 VAC 5-80-1210)

18. Permit Suspension/Revocation - This permit may be suspended or revoked if the permittee:

- a. Knowingly makes material misstatements in the permit application or any amendments to it;
- b. Fails to comply with the conditions of this permit;
- c. Fails to comply with any emission standards applicable to a permitted emissions unit;
- d. Causes emissions from the stationary source which result in violations of , or interfere with the attainment and maintenance of, any ambient air quality standard; or
- e. Fails to operate in conformance with any applicable control strategy, including any emission standards or emission limitations, in the State Implementation Plan in effect at the time an application for this permit is submitted.

(9 VAC 5-80-1210 G)

19. Right of Entry - The permittee shall allow authorized local, state, and federal representatives, upon the presentation of credentials:

- a. To enter upon the permittee's premises on which the facility is located or in which any records are required to be kept under the terms and conditions of this permit;
- b. To have access to and copy at reasonable times any records required to be kept under the terms and conditions of this permit or the State Air Pollution Control Board Regulations;
- c. To inspect at reasonable times any facility, equipment, or process subject to the terms and conditions of this permit or the State Air Pollution Control Board Regulations; and
- d. To sample or test at reasonable times.

For purposes of this condition, the time for inspection shall be deemed reasonable during regular business hours or whenever the facility is in operation. Nothing contained herein shall make an inspection time unreasonable during an emergency.

(9 VAC 5-170-130 and 9 VAC 5-80-1180)

20. Maintenance/Operating Procedures – At all times, including periods of start-up, shutdown, and malfunction, the permittee shall, to the extent practicable, maintain and operate the

affected source, including associated air pollution control equipment, in a manner consistent with good air pollution control practices for minimizing emissions.

The permittee shall take the following measures in order to minimize the duration and frequency of excess emissions, with respect to air pollution control equipment and process equipment that affect such emissions:

- a. Develop a maintenance schedule and maintain records of all scheduled and non-scheduled maintenance.
- b. Maintain an inventory of spare parts.
- c. Have available written operating procedures for equipment. These procedures shall be based on the manufacturer's recommendations, at a minimum.
- d. Train operators in the proper operation of all such equipment and familiarize the operators with the written operating procedures, prior to their first operation of such equipment. The permittee shall maintain records of the training provided including the names of trainees, the date of training and the nature of the training.

Records of maintenance and training shall be maintained on site for a period of five years and shall be made available to DEQ personnel upon request.

(9 VAC 5-50-20 E and 9 VAC 5-80-1180 D)

21. **Record of Malfunctions** – The permittee shall maintain records of the occurrence and duration of any bypass, malfunction, shutdown or failure of the facility or its associated air pollution control equipment that results in excess emissions for more than one hour. Records shall include the date, time, duration, description (emission unit, pollutant affected, cause), corrective action, preventive measures taken and name of person generating the record.
(9VAC 5-20-180 J and 9 VAC 5-80-1180 D)
22. **Notification for Facility or Control Equipment Malfunction** - The permittee shall furnish notification to the Director, Piedmont Region of malfunctions of the affected facility or related air pollution control equipment that may cause excess emissions for more than one hour, by facsimile transmission, telephone or telegraph. Such notification shall be made as soon as practicable but no later than four daytime business hours after the malfunction is discovered. The permittee shall provide a written statement giving all pertinent facts, including the estimated duration of the breakdown, within two weeks of discovery of the malfunction. When the condition causing the failure or malfunction has been corrected and the equipment is again in operation, the permittee shall notify the Director, Piedmont Region.
(9 VAC 5-20-180 C and 9 VAC 5-80-1180)
23. **Violation of Ambient Air Quality Standard** - The permittee shall, upon request of the DEQ, reduce the level of operation or shut down a facility, as necessary to avoid violating

any primary ambient air quality standard and shall not return to normal operation until such time as the ambient air quality standard will not be violated.

(9 VAC 5-20-180 I and 9 VAC 5-80-1180)

24. **Change of Ownership** - In the case of a transfer of ownership of a stationary source, the new owner shall abide by any current permit issued to the previous owner. The new owner shall notify the Director, Piedmont Region of the change of ownership within 30 days of the transfer.

(9 VAC 5-80-1240)

25. **Permit Copy** - The permittee shall keep a copy of this permit on the premises of the facility to which it applies.

(9 VAC 5-80-1180)